Unit 48 Green chemistry

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

48.1 Sustainable development and green chemistry

During the twentieth century, the chemical industry has fundamentally changed the way we live, including the cars we drive, the clothes we wear, the food we eat and the everyday conveniences that we take for granted.

However, the chemical industry is often viewed as causing more harm than good. A major reason is that the industry is perceived as one which causes significant environmental damage. The challenge for the chemical industry is to continue to provide the benefits that we rely on, in an economically viable manner, but without the adverse environmental side effects.



Fig. 48.1 Air pollutants from industry

According to the United Nations,

> Sustainable development can be defined as 'development meeting the needs of the present without compromising the ability of future generations to meet their needs'.

The three components of sustainable development are economic growth, environmental protection and social equity. It is suggested that we can achieve all three by gradually changing the ways in which we develop and use technologies.



Fig. 48.2 The three components of sustainable development

Historically, the general public was not willing to place a high priority on protecting the environment when there was concern about achieving a certain level of welfare or economic goals. Now for many nations, the ability of the economy to meet basic needs allows them to focus more on environmental issues.

As governments and businesses focus on the transition to sustainability, they increasingly recognize the need to balance the often conflicting challenges of economic growth, environmental protection and social progress.

Green chemistry is chemistry for the environment. It was first N1 defined by the United States in the early 1990s as 'the utilization of a set of principles that reduce or eliminate the use or generation of hazardous substances in the design, manufacture and use of chemical products'. Green chemistry has gradually become recognized as a method for achieving sustainability.

N1 Paul Anastas — the 'father of green chemistry'
Paul Anastas is an organic chemist and the man widely credited with starting the grassroots movement to change the way chemists work.

Paul Anastas summarized his philosophy in his '12 principles of green chemistry'. Since 1996, an annual Presidential Green Chemistry Challenge has been held in the US. Prizes are awarded to companies that have developed greener processes, reaction conditions or products.

green chemistry 綠色化學

12 principles of green chemistry

1 Prevention — It is better to prevent waste than to treat or clean up after waste has been formed.

2 Atom economy — When creating products it is important to maximize the incorporation of all the materials used into the final product.

ASL 2009 Paper 2 Q5(c)
ASL 2010 Paper 1 Q11
AL 2009 Paper 2 Q4(c)(iii)

ASL 2003 Q12(a)

3 Less hazardous synthesis — Processes should be designed to use and generate substances that have little to no toxicity.

4 Designing safer chemicals — Chemical products should be designed to function well while reducing toxicity.

5 Safer auxiliary substances — Whenever possible, avoid the use of auxiliary substances such as solvents and separating agents. When these substances are needed, they should be non-toxic. N2

ASL 2009 Paper 2 Q5(c)
ASL 2010 Paper 1 Q11
AL 2009 Paper 2 Q4(c)(iii)

N2 Refer to page T55.

6 Energy efficiency — Energy efficiency should be considered when designing and producing a product.

7 Use of renewable resources — When possible, the feedstocks should come from a renewable resource.

8 Reducing derivatives — Unnecessary derivatization (e.g. use of blocking groups) should be avoided if possible.

ASL 2010 Paper 1 Q11

9 Catalysis — Catalysts speed up chemical processes and reduce the energy requirement of processes. N3

N3 Catalysts are important in the development of greener chemical synthesis.

- Catalysts affect the conditions that are needed for chemical processes, often reducing the time for production and saving energy.
- Many catalysts are also highly selective, converting a high percentage of the consumed reactant to the desired product. Few or no by-products are formed, thus reducing the waste.
- Biocatalysts, enzymes produced by microorganisms, are able to catalyze chemical processes efficiently under mild conditions.

10 Design for degradation — Design chemicals and products that can degrade after use.

ASL 2009 Paper 2 Q5(c)
AL 2009 Paper 2 Q4(c)(iii)

11 Use real-time analysis for pollution prevention — Methods need to be further developed to allow for real-time, in-process monitoring and control prior to the formation of hazardous substances.

12 Accident prevention — Substances and their states used in chemical processes should be chosen to minimize the potential for chemical accidents, including fires, explosions and releases.

Fig. 48.3 12 principles of green chemistry

AL 2008 Paper 1 Q10

We may regard green chemistry as a 'reduction' process which aims to reduce materials, energy, risk and hazards, waste, cost and environmental impact, etc.

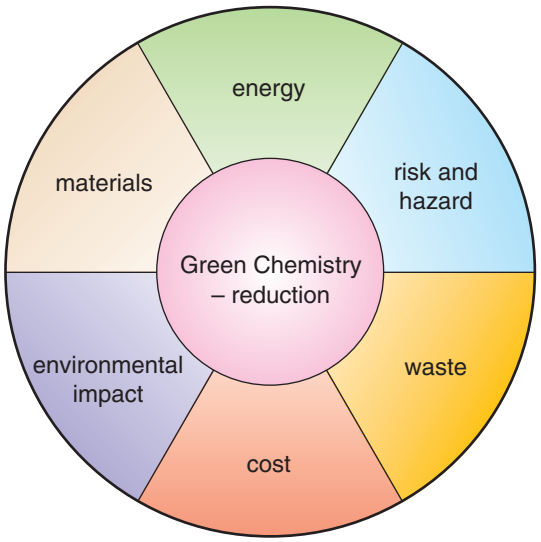


Fig. 48.4 Green chemistry can be regarded as a reduction process

Environmental, economic and social concerns must be addressed at the same time. For example, if we fail to protect the environment, we will have no jobs, no communities, and no future; but if we attempt to impose environmental solutions that ignore economic and social realities, we will face disaster of a different sort.

In the following sections, we will look at some innovations in the green chemistry, including

- feedstocks from renewable resources;
- atom economy; and
- energy efficiency.

48.2 Feedstocks from renewable resources

One of the principles of green chemistry is to use renewable resources. At present, petroleum is the main source of chemical feedstocks. Reducing the reliance on petrochemicals will help to make the industry more sustainable.

* We can grow corn year after year and thus it is a renewable resource.

Currently most polymers are made from petrochemicals. DuPont company has developed a way of making one of the two monomers for the production of the polymer Sorona from corn*.



Fig. 48.5 DuPont company makes one of the two monomers for the production of the polymer Sorona from corn

The method relies on fermenting glucose extracted from corn into malonic acid, followed by catalytic hydrogenation to propane-1,3-diol (PDO). PDO is then co-polymerized with terephthalic acid (TPA) to form Sorona. Manufacturers convert Sorona into fibres for clothing and upholstery. Fig. 48.6 summarizes the above steps.

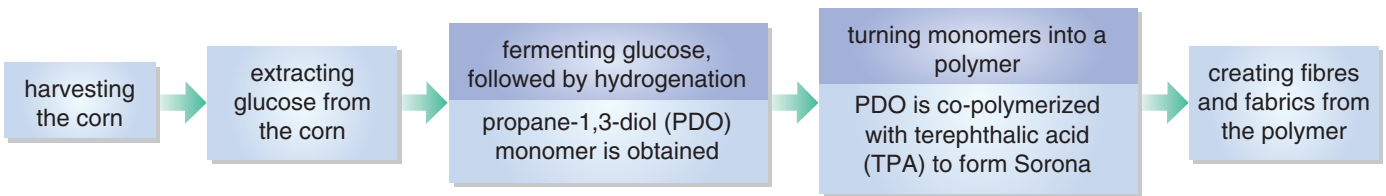


Fig. 48.6 A summary of production of the polymer Sorona from corn

The development of Sorona illustrates the practices of green chemistry principles in chemical industry, including:

- the use of feedstock from a renewable resource;
- the production of one monomer happens at low temperature with enzymes.

AL 2008 Paper 1 Q10

* We have discussed calculating the percentage yield in Topic 3 Metals.

48.3 Atom economy

Most chemists have traditionally measured the efficiency of a reaction by the percentage yield*.

$$\text{Percentage yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

However, this only tells part of the story.

Suppose we want to produce 1-bromobutane using the following substitution reaction:

$$C_4H_9OH(l) + NaBr(s) + H_2SO_4(aq) \longrightarrow C_4H_9Br(l) + NaHSO_4(aq) + H_2O(l)$$

In a certain experiment, 0.800 g of butan-1-ol, 1.33 g of sodium bromide, and 2.00 g of concentrated sulphuric acid were allowed to react. 1.20 g of 1-bromobutane were obtained.

Table 48.1 shows the number of moles of reagents used.

Table 48.1 Number of moles of reagents used in the above reaction

Reagent	Molar mass ($g\ mol^{-1}$)	Mass present (g)	Number of moles present (mol)
C_4H_9OH	74.0	0.800	0.0108
NaBr	102.9	1.33	0.0129
H_2SO_4	98.1	2.00	0.0204

According to the equation, 1 mole of C_4H_9OH reacts with 1 mole of NaBr and 1 mole of H_2SO_4. Therefore C_4H_9OH is the limiting reagent.

$$\text{Theoretical yield of } C_4H_9Br = 0.0108 \text{ mol} \times 136.9 \text{ g mol}^{-1} = 1.48 \text{ g}$$

$$\text{Percentage yield of } C_4H_9Br = \frac{1.20 \text{ g}}{1.48 \text{ g}} \times 100\% = 81.1\%$$

In the above experiment, 4.13 g of reactants (0.800 g of butan-1-ol, 1.33 g of NaBr and 2.00 g of H_2SO_4) were used, and that even with a 100% yield, this reaction would only yield 1.48 g of the desired product.

The bulk (4.13 g – 1.48 g = 2.65 g) of the mass of reactants ends up in by-products ($NaHSO_4$ and H_2O).



Fig. 48.7 Barry Trost

Even if the reaction has 100% yield, only 35.8% ($\frac{1.48\ g}{4.13\ g}$ x 100%) of the mass of the reactants end up in the desired product. Now the actual yield of the reaction is 81.1%, then only 29.0% (0.811 x 0.358 x 100%) of the mass of the reactants actually ends up in the desired product.

To foster awareness of atoms of reactants that are incorporated into the desired product and those that are wasted, Barry Trost, a professor of chemistry at Stanford University, developed the concept of **atom economy**. Let us consider the above substitution reaction again in light of the concept of atom economy.

We can rewrite the equation in the above example, showing all the reactant atoms that are incorporated into the desired product in green and those that are wasted in brown.

$$C_4H_9—OH(l) + Na—Br(s) + H_2SO_4(aq) \longrightarrow C_4H_9—Br(l) + NaHSO_4(aq) + H_2O(l)$$

Table 48.2 provides another view of this reaction based on the concept of atom economy. The atoms of the reactants that are incorporated into the desired product are shown in green (column 2) while atoms of the reactants that end up in by-products are shown in brown (column 3).

Focusing on the last row of this table, we can see that of all the atoms of the reactants, only 4C, 9H, and 1Br are incorporated into the desired product and the bulk (3H, 5O, 1Na, 1S) are wasted. This is an example of poor atom economy.

Table 48.2 Looking at the substitution reaction based on the concept of atom economy

Formula of reagent	Atoms incorporated into desired product	Atoms in by-products
C_4H_9OH	4C, 9H	1H, 1O
NaBr	1Br	1Na
H_2SO_4	—	2H, 4O, 1S
Total 4C, 12H, 5O, 1Br, 1Na, 1S	**Total** 4C, 9H, 1Br	**Total** 3H, 5O, 1Na, 1S

atom economy 原子經濟

A logical extension of the concept of atom economy is to calculate the atom economy. This can be done by taking the ratio of the relative molecular mass or formula mass of the desired product to the total relative molecular mass or formula mass of the reactants and multiplying by 100%.

$$\text{Atom economy} = \frac{\text{relative molecular mass or formula mass of desired product}}{\text{total relative molecular mass or formula mass of reactants}} \times 100\%$$

N4 The calculation of atom economy is based on a 100% completed reaction. Most reactions do not go to completion and the yield is related to the extent of the reaction. Thus, a reaction with a high atom economy does not necessarily have a high yield.

N4

ASL 2009 Paper 2 Q5(a)
AL 2009 Paper 2 Q4(c)(i)
DSE 2013 Paper 2 Q1(c)(ii)

	$C_4H_9OH(l)$	+ $NaBr(s)$	+ $H_2SO_4(aq)$	$\longrightarrow$	$C_4H_9Br(l)$	+ $NaHSO_4(aq)$	+ $H_2O(l)$
relative molecular mass or formula mass	74.0	102.9	98.1		136.9		

Atom economy of the substitution reaction

$$= \frac{136.9}{74.0 + 102.9 + 98.1} \times 100\%$$

$$= 49.8\%$$

N5 The Haber process as an example of green chemistry

The Haber process is an energy-intensive process, as a large amount of energy is required for attaining the high temperature and pressure for accomplishing the reaction. However, the Haber process can be considered as an example of green chemistry because

- the reaction has 100% atom economy;
- the reactants (N_2 and H_2) are non-toxic and pose no harm to the environment;
- the reaction takes place in gas phase (no solvent is used);
- the product is separated by cooling (no separating agent is used);
- the raw material (N_2) is available in large amount in the atmosphere (depletion is not a problem);
- the reaction does not require the use of any derivatives;
- a catalyst is used in the process.

The atom economy of the reaction is 49.8%. Thus, even for 100% yield, only half of the mass of the reactants will be incorporated into the desired product while the rest will be wasted in by-products. This is not a green process.

One of the key principles of green chemistry is that processes should be designed to incorporate the maximum amount of all the reactants into the final product, thus reducing waste products.

A greener pain reliever N5 N6

Ibuprofen is a popular pain reliever.

Boots patented ibuprofen in the 1960s. At that time there were six stages in the complex process to make the drug.



Fig. 48.8 Ibuprofen

N6 Refer to page T56.

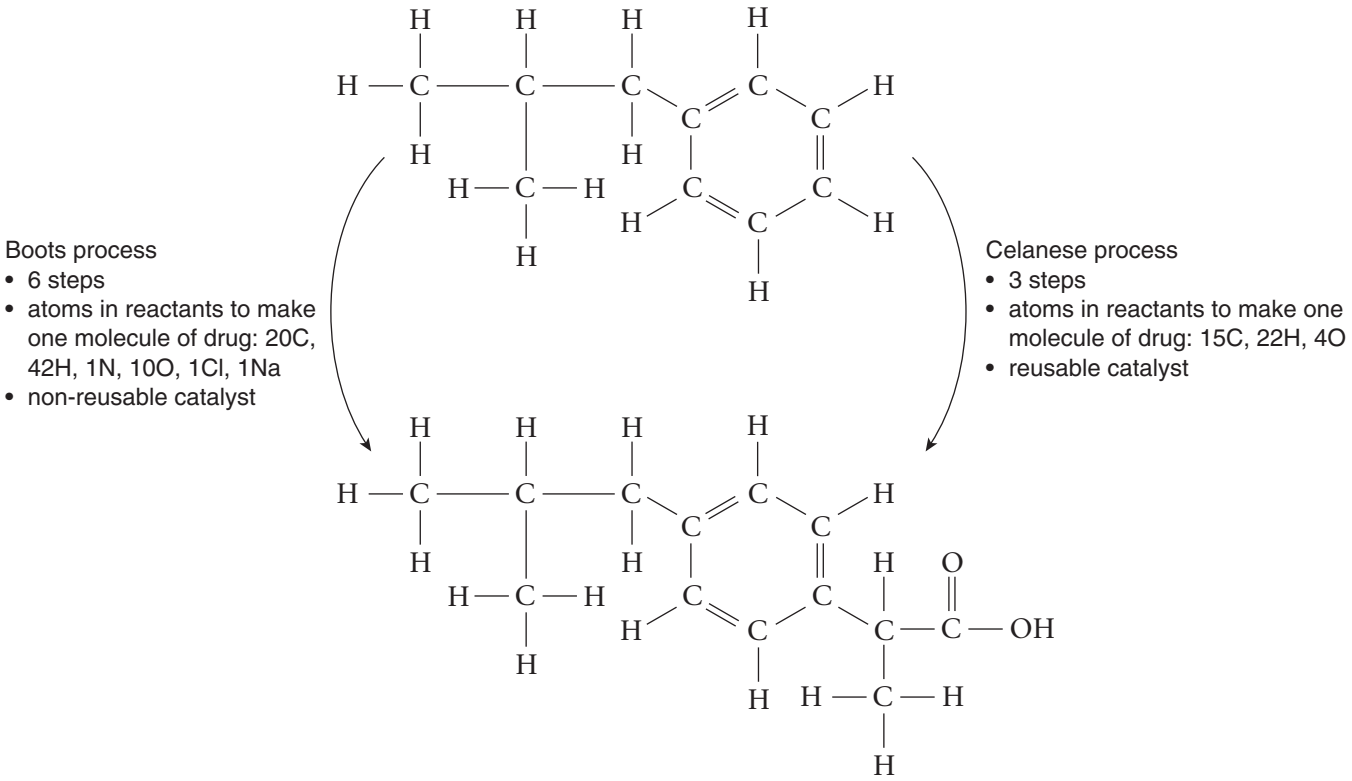


Fig. 48.9 Two routes for making ibuprofen from the same starting material

There are 33 atoms of all kinds in one ibuprofen molecule. The Boots process needed 75 atoms for each ibuprofen molecule formed (Fig. 48.9). So 42 atoms were wasted.

Once the Boots patent for ibuprofen ran out after 20 years, it was possible for other companies to make and sell the drug. A new synthesis was developed by the Celanese Corporation which was more efficient. This new process has only three stages and needs only 41 atoms in the reacting chemicals to make each ibuprofen molecule (Fig. 48.9). It has a much more efficient atom economy. It gives fewer harmful by-products. It is a greener synthesis.

Another feature of the Boots process was that it used aluminium chloride as a catalyst in one key step of the process. Although a catalyst, the aluminium chloride could not be recovered and reused. This added to the waste.

The newer process uses other catalysts: hydrogen fluoride and an alloy of nickel and aluminium. Both of these catalysts can be recovered and reused many times.

AL 2008 Paper 1 Q10



48.1 Comparing the atom economy of two reactions

a) An amide, $CH_3CH_2CONHCH_3$, can be prepared from ethyl propanoate according to the following equation:

$$CH_3CH_2COOCH_2CH_3 + CH_3NH_2 \longrightarrow CH_3CH_2CONHCH_3 + CH_3CH_2OH$$

Calculate the atom economy of the reaction.

b) The same amide can also be prepared using methyl propanoate instead of ethyl propanoate:

$$CH_3CH_2COOCH_3 + CH_3NH_2 \longrightarrow CH_3CH_2CONHCH_3 + CH_3OH$$

Calculate the atom economy of this reaction.

(Relative atomic masses: H = 1.0, C = 12.0, N = 14.0, O = 16.0)

Solution

a)

	$CH_3CH_2COOCH_2CH_3$	+ CH_3NH_2	$\longrightarrow$	$CH_3CH_2CONHCH_3$	+ CH_3CH_2OH
relative molecular mass	102.0	31.0		87.0	

$$\text{Atom economy} = \frac{\text{relative molecular mass or formula mass of desired product}}{\text{total relative molecular mass or formula mass of reactants}} \times 100\%$$

$$= \frac{87.0}{102.0 + 31.0} \times 100\%$$

$$= 65.4\%$$

b)

	$CH_3CH_2COOCH_3$	+ CH_3NH_2	$\longrightarrow$	$CH_3CH_2CONHCH_3$	+ CH_3OH
relative molecular mass	88.0	31.0		87.0	

$$\text{Atom economy} = \frac{\text{relative molecular mass or formula mass of desired product}}{\text{total relative molecular mass or formula mass of reactants}} \times 100\%$$

$$= \frac{87.0}{88.0 + 31.0} \times 100\%$$

$$= 73.1\%$$

$\therefore$ the atom economy of the first reaction is 65.4% while that of the second reaction is 73.1%.

From the above example, we can see that a small change in the design of the synthesis can lead to a significant improvement in the atom economy.



Checkpoint

Refer to page T57.

Ethyl ethanoate can be prepared in the following way:

$$CH_3COCl + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + HCl$$

a) Calculate the atom economy of this reaction.
70.7%

b) Ethyl ethanoate can be prepared from the reaction between ethanoic acid and ethanol.

i) Write an equation for this reaction.

ii) Work out the atom economy of this reaction.
83.0%

iii) The atom economy of this reaction is greater than that calculated in (a).

Explain why this reaction has a greater atom economy.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0, Cl = 35.5)

48.4 Energy efficiency

Burning natural gas or other fossil fuels is the usual source of heat in the chemical industry. Often the heat from burning is used to produce super-heated steam, which can then be used for heating around the chemical plant.

The most direct way of reducing the use of heat is to prevent loss of steam from leaking valves on steam pipes and by installing efficient insulation on reaction vessels or pipes.

Some of the reactions in chemical processes may be so exothermic that they provide the heat to raise steam and generate electricity which can be 'exported' to other processes nearby.

The aim of green chemistry is not only to make processes more energy efficient but also to lower their energy demand. One of the targets of current research is to develop new processes that run at much lower temperatures. One way of doing this is to use biocatalysts – enzymes*.

* We have discussed this in Unit 46.

48.5 Manufacture of acetic acid in industry

Acetic acid (ethanoic acid) is one of the world's most important chemicals and serves as an intermediate in the production of a vast range of products (Fig. 48.10).

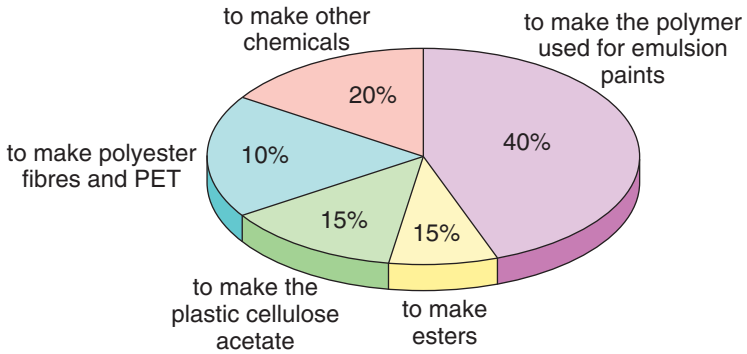


Fig. 48.10 The uses of acetic acid

Processes that have been developed to manufacture acetic acid illustrate the application of principles of green chemistry.

Oxidation of naphtha or butane

Up to the 1970s, acetic acid was produced industrially by the direct oxidation of naphtha or butane, and this process is still in operation in some countries.

The process operates at 180 – 200 °C and 40 – 50 atmospheres with a cobalt(II) ethanoate catalyst. The naphtha and butane are derived from petroleum. It produces a wide range of products (Fig. 48.11). The cost of separation of these products by fractional distillation is offset by the fact that these products can be sold.

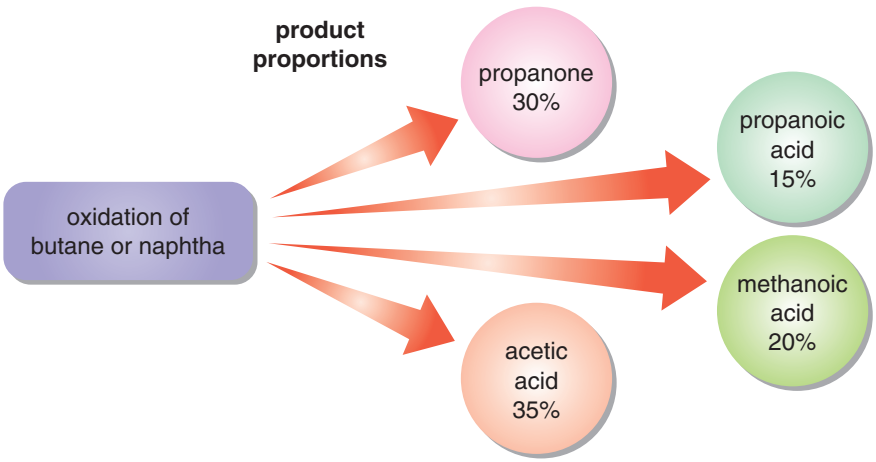


Fig. 48.11 Oxidation of butane or naphtha produces a wide range of products

The Monsanto process

From about 1970, acetic acid was mainly produced by a new process developed by Monsanto. In this process, methanol reacts with carbon monoxide in the presence of a catalyst made of rhodium metal and iodide ion.

$$CH_3OH(l) + CO(g) \xrightarrow[\substack{150 - 200\ ^\circ C,\\ 30 - 60 \text{ atmospheres}}]{\text{rhodium / iodide ion}} CH_3COOH(l) \qquad \Delta H = -135 \text{ kJ mol}^{-1}$$

This process was first commercialized in 1970 and was a great advance on the previous processes in several aspects:

- It has an atom economy of 100%. In contrast, the oxidation of naphtha or butane has a very poor atom economy (about 35%). Consequently, there is a huge reduction in waste.
- The whole process uses less energy, particularly for product separation and purification.
- The yield of acetic acid is high. About 98% of the methanol is converted to the acid.
- Methanol is a cheaper feedstock when compared with naphtha or butane.
- Methanol* can be produced from biomass, municipal wastes and sewage.
- The reaction is extremely fast and the catalyst has a long life.

The Monsanto process was the dominant process from the 1970s to the late 1990s. N7

* At present, methanol is mainly manufactured from natural gas.

N7 Disadvantages with the Monsanto process

Disadvantages with the Monsanto process include

- rhodium is very expensive;
- rhodium and iodide ions form insoluble salts and the water content in the reaction vessel has to be kept relatively high to prevent this;
- a final distillation step is required to remove water, adding to the energy and costs demand;
- rhodium also catalyzes side reactions.

The Cativa process

In 1986, the oil company BP Chemicals Limited acquired the Monsanto acetic acid technology and used iridium metal and iodide ion as a catalyst. The result was the Cativa process.

The atom economy for the Cativa process is also 100%. The use of iridium-based catalyst has numerous benefits:

- Iridium costs only about one fifth as much as rhodium.
- The process is faster and smaller reactors can be used.

* The selectivity of a catalyst indicates the percentage of consumed reactant that is converted to the desired product.

- Iridium is even more selective* for methanol, which increases the overall yield and reduces by-products, resulting in lower purification costs and reduced waste.
- The water content in the reaction vessel can be reduced, thus reducing the energy needed at the distillation and purifying stages.
- Existing plants can be converted to run the Cativa process. A converted plant can produce up to 75% more acetic acid than was previously possible using the Monsanto process.

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website.*



Discussion

Here you are going to study the Solvay process and discuss whether the process meets the principles of green chemistry.

The Solvay process

Sodium hydrogencarbonate and sodium carbonate are industrial chemicals required for glass making, cleaning formulations and many other applications. These two chemicals are made by the Solvay process.

The key reaction in the process involves saturating a sodium chloride solution (brine) with ammonia and carbon dioxide, followed by cooling. Solid sodium hydrogencarbonate comes out of the solution.

$NaCl(aq) + NH_3(g) + CO_2(g) + H_2O(l) \longrightarrow NaHCO_3(s) + NH_4Cl(aq)$ ..... (1)

When heated, the solid sodium hydrogencarbonate yields solid sodium carbonate, carbon dioxide and water.

sodium hydrogencarbonate $\xrightarrow{\text{heat}}$ _____ ..... (2)

The carbon dioxide produced is recycled back for use in reaction (1).

In addition to sodium chloride, the major consumable raw material is calcium carbonate. Calcium carbonate is heated to produce calcium oxide and carbon dioxide.

calcium carbonate $\xrightarrow{\text{heat}}$ _____ ..... (3)

The carbon dioxide is used in reaction (1). The calcium oxide is reacted with water to produce calcium hydroxide. Calcium hydroxide is then heated with the ammonium chloride in the solution from reaction (1). The ammonia released is recycled back for use in reaction (1).

calcium hydroxide + ammonium chloride $\xrightarrow{\text{heat}}$ _____ ..... (4)

1 Use chemical equations to describe reactions (2), (3) and (4).

2 Which aspects of the Solvay process meet the principles of green chemistry?

3 Suggest how the process can be made greener.

Extension

Discussion

1 Equation (2)
$2NaHCO_3(s) \longrightarrow Na_2CO_3(s) + CO_2(g) + H_2O(l)$
Equation (3)
$CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$
Equation (4)
$Ca(OH)_2(s) + 2NH_4Cl(aq) \longrightarrow CaCl_2(aq) + 2NH_3(g) + 2H_2O(l)$

2
- It uses inexpensive, abundantly available raw materials (sodium chloride and calcium carbonate). The most expensive raw material is ammonia. However, as ammonia is recycled, only a relatively small quantity of additional ammonia is required.
- It recycles two major reactants, ammonia and carbon dioxide.

3 The Solvay process requires significant amount of energy. Using energy from renewable resources can make the process greener.

48.6 Green chemistry and the life cycle of a product

Fig. 48.12 shows the stages of the life cycle of a product. So far green chemistry innovations mainly focus on the materials and manufacturing stages of the life cycle of a product. Actually green chemistry can facilitate environmental improvements at every stage of the life cycle.

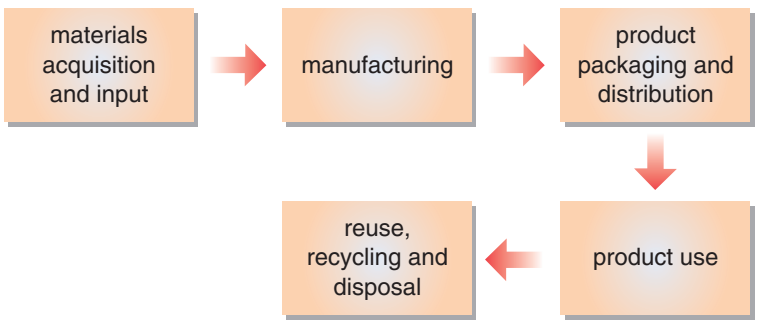


Fig. 48.12 Stages of the life cycle of a product

Consider packaging and distribution, an important element in terms of environmental impact. Excessive packaging consumes resources and increases the waste disposal problem. There are examples of products that are designed to completely eliminate the need for packaging. For example, in recent years software has moved from excessive packaging to no packaging. Software and manuals are downloaded directly from the internet.

Extension



Fig. 48.13 A 'concentrated' fabric softener

In recent years, 'concentrated' fabric softeners have been introduced; the same volume of softener can be used for more loads of laundry. Such an improvement decreases the packaging required and the product transportation cost.

Product use may also have an impact on the environment. The product itself, such as insecticides or chemicals used in dry cleaning, may be potentially hazardous to human and environmental health. These impacts are minimized through the application of green chemistry.

Environmentally friendly packaging of products

Reducing the packaging of products can reduce budget and address the environmental protection objective of a company.

Recently, the largest consumer goods production company in the United States, P&G, has been advocating environmentally friendly packaging of products. They announced that their products, Crest (toothpaste) and Tide (detergent), would be repackaged. A harder plastic material would be selected for the surface of the newly packaged toothpaste. In addition, the circumference of the lid for the toothpaste tube would be enlarged so that it can 'stand upside down' without relying on the support of the packaging carton. On the other hand, although the volume of the new style detergent would shrink, the concentration would be doubled.

The Chief Financial Officer of the company stated that new packages would inevitably boost the production costs. New moulds for producing the package would be needed, and expenses for promoting sales would also increase. He expected the new packaging scheme could create a win-win situation for consumers, retailers, the environment and the company.

The largest retailer of the world, Wal-Mart, is currently persuading 66 000 of their producers to give up over-packaging. The company committed that all the consumer goods in their shops will conform to the principle of neutral packaging before 2025, which means the materials used for packaging being recyclable or degradable.

However, the recycling industry is still confronting some difficulties. The inadequacy of support from the government for recycling materials results in the significant increase in the production cost, and thus causes producers to abstain from recycling materials. Meanwhile, the pursuit of eye-catching packages from consumers also leads producers to avoid using over-simplified packaging designs.

(Source: JKP 06/2008)

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

Refer to page T58.

Green chemistry does not just guide the work of chemists and manufacturers, it can be applied in various aspects of our lives.

1 Make a list of 10 ways of how you can make your own living greener.

2 Try to practise some of the suggested ways for one week and share your experience in the class.

You may visit the following website for some ideas:
http://eartheasy.com/homepage.htm

Worksheet
Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

STSE Connections

Greener dry cleaning

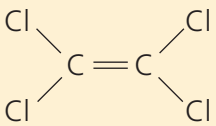
Whenever we purchase an article of clothing whose tag says 'Dry Clean Only', we know we are going to pay more to keep that garment clean. But have we ever considered the environmental cost associated with dry cleaning?



Fig. 48.14 An article of clothing that needs dry cleaning

The solvent commonly used in dry cleaning

The term 'dry cleaning' is a little misleading since it is a liquid chemical solvent that removes the dirt and stains from clothes. However, instead of water, the solvent used by most dry cleaners is tetrachloroethene.



Tetrachloroethene has serious environmental and health impacts. It is toxic to the liver and the central nervous system. It can accumulate in the body and is probably carcinogenic to humans. Tetrachloroethene is very persistent in ground water and soil, as well as toxic to the aquatic environment.



Fig. 48.15 The use of tetrachloroethene in dry cleaning raises health concerns for workers in dry cleaning companies

Liquid carbon dioxide — a greener alternative

The environmental problems associated with dry cleaning clothes using tetrachloroethene and the rising costs of complying with environmental laws encourage the search for alternative methods.

One promising new approach to cleaning clothes is using liquid carbon dioxide* instead of tetrachloroethene.

* Actually supercritical carbon dioxide is used. We will discuss supercritical carbon dioxide in Topic 15 Materials Chemistry.

One of the biggest challenges in cleaning clothes is the removal of stains, such as grease and oil. We cannot remove greasy stains by rinsing the clothes in plain water. Detergent is needed to complete the job. A detergent particle contains a polar head that is soluble in water and a non-polar tail that dissolves in grease or oil.

Unfortunately, even though CO_2 is non-polar, it does not attract grease or oil strongly enough to remove them from the clothes. This problem initially restricted applications of CO_2 in the dry cleaning industry. A researcher discovers a group of polymers that can act as detergents in this environmentally benign solvent.

Cleaning clothes with liquid CO_2 offers several advantages over traditional dry cleaning processes. Tetrachloroethene is eliminated as a solvent, removing the human health concerns associated with the use of this substance. Liquid CO_2 is a non-flammable, non-toxic substance. Although CO_2 is a greenhouse gas, no new CO_2 is generated for use in this technology. CO_2 that is produced during fossil fuel combustion and other industrial processes may be used.

In a comparison with tetrachloroethene, consumer reports suggest that CO_2 offers better results in tests for shrinkage, discoloration and preservation of texture.

Questions

1 It is suggested that after collecting the clothes from a dry cleaner, we should remove the plastic bag and hang the clothes in a well-ventilated place for a few days before putting them in the wardrobe. Suggest a possible reason for this action.

2 Explain how a detergent solution cleans a piece of cloth with a greasy stain.

3 Despite the benefits offered by dry cleaning using liquid CO_2, few laundries have adopted this new technology at present. Suggest possible reasons for this.

STSE Connections

1 When dry-cleaned clothes are brought home, they continue to give off solvent residue in the form of gas.
So, remove the plastic bag and allow clothes to air outside before putting them in the wardrobe, avoiding the clothes from emitting gas in the room where you sleep.

2 When a detergent solution is added to a cloth with a greasy stain, the hydrophobic 'tails' of the detergent particles dissolve in the grease.
The hydrophilic 'heads' of the detergent particles are insoluble in the grease and remain outside.
The surrounding water molecules attract the anionic 'heads' and lift the grease off the surface.
By stirring, the grease (which carries the dirt particles) breaks up into tiny droplets suspended in the water. These tiny droplets cannot come together again due to the repulsion between the hydrophilic 'heads' of detergent particles on their surfaces. An emulsion forms.
Rinsing washes away the greasy suspension, leaving the surface clean.

3 The majority of dry cleaners continue to use tetrachloroethene because there is little awareness and no legislative imperative to choose safer substitutes.



Worksheet Assessment Form

A worksheet and an assessment form on this activity are available for download from *Jing Kung Chemistry Website*.

Decision Making

Refer to page T58.

A chemical plant in City X?

Let us come back to the case of building a chemical plant in City X mentioned at the beginning of Unit 47. Remember that three companies are interested in building chemical plants in the city:

- AA Nitrogenous Fertilizer Company
- BB Chlorine Gas Company
- CC Methanol Company

The economy of the city has gone down since the bankruptcy of a local bank and many residents have lost their jobs as so many companies have closed.

The city council meeting to discuss the possibility of a chemical plant in City X will be held soon. Representatives of the three companies and all interested citizens will attend.

Tasks

Your teacher will divide the class into groups.

Groups as company representatives

Three groups will act as representatives of the companies. Each group will give a 4-minute presentation on the following aspects in the meeting:

1 production technologies the plant use;

2 economic and social benefits of their industry to City X;

3 measures to be taken to minimize environmental impacts of the plant.

Other groups

Other groups will be citizens attending the meeting. You need to

1 based on the principles of green chemistry, develop at least six expectations that the companies should meet if they choose to locate in City X;

2 raise two questions in the meeting based on the list of expectations;

3 decide whether a chemical plant should be built in the city and justify your decision.

Summary

1 Sustainable development can be defined as 'development meeting the needs of the present without compromising the ability of future generations to meet their needs'.

2 Green chemistry is 'the utilization of a set of principles that reduce or eliminate the use or generation of hazardous substances in the design, manufacture and use of chemical products'.

3 We may regard green chemistry as a 'reduction' process which aims to reduce materials, energy, risk and hazards, waste, cost and environmental impact, etc.

4 Atom economy

$$= \frac{\text{relative molecular mass or formula mass of desired product}}{\text{total relative molecular mass or formula mass of reactants}} \times 100\%$$

5 Processes that have been developed to manufacture acetic acid illustrate the application of principles of green chemistry. These processes include:

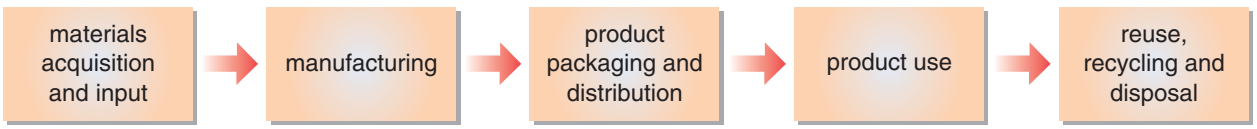
a) oxidation of naphtha or butane;

b) the Monsanto process;

c) the Cativa process.

Extension

6 The following diagram shows the stages of the life cycle of a product. Green chemistry can facilitate environmental improvements at every stage of the life cycle.



Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 a) Explain the meaning of the term 'sustainable development'. Sustainable development can be defined as 'development meeting the needs of the present without compromising the ability of future generations to meet their needs'.

b) What are the THREE components to sustainable development?
Economic growth, environmental protection and social equity

2 Give FOUR principles of green chemistry.
Refer to page T59.

3 Construct a concept map for concepts learnt in this unit.
Refer to page T60.

Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

4 For each of the following green chemistry practices, suggest ONE example from the chemical industry.

a) Managing energy inputs and outputs The reaction between nitrogen and hydrogen is exothermic. In the Haber process, the hot product gases are used to warm up the nitrogen and hydrogen in a heat exchanger before they enter the catalytic converter.

b) Use of renewable resources
Biomass is used to produce syngas for methanol synthesis.

c) Recycling chemicals The reaction between nitrogen and hydrogen is reversible. In the Haber process, the percentage of ammonia in the equilibrium mixture is about 30 – 40%. The unreacted nitrogen and hydrogen are recycled.

d) Reducing waste products
In the Haber process, the unreacted nitrogen and hydrogen are recycled so as to reduce waste products.

5 Governments regulate chemical companies. For each of the following areas, suggest a reason why it is important to have rules to protect the public, people at work, or the environment.

a) Choice of raw materials for manufacturing
Using highly hazardous raw materials for manufacturing poses hazards to people working in the plant and the environment.

b) Transporting chemicals
Any accident that occurs during the transport of chemicals may pose hazards to the public and the environment.

c) Storing chemicals Improper storage of chemicals may lead to spillage, posing hazards to people working in the plant, the public and the environment.

d) Getting rid of chemical waste
Discharging untreated chemical waste into the water poses hazards to the public and the environment.

6 The following two methods can be used to convert copper metal into copper(II) nitrate solution:

Method 1: $Cu(s) \longrightarrow CuO(s) \xrightarrow{\text{dilute } HNO_3(aq)} Cu(NO_3)_2(aq)$

Method 2: $Cu(s) \xrightarrow{\text{dilute } HNO_3(aq)} Cu(NO_3)_2(aq)$

a) Refer to *Method 1*.

i) Suggest how copper metal can be converted into copper(II) oxide. State the expected observation in the reaction that you have suggested.

ii) Name the type of reaction that occurs between copper(II) oxide and dilute nitric acid.

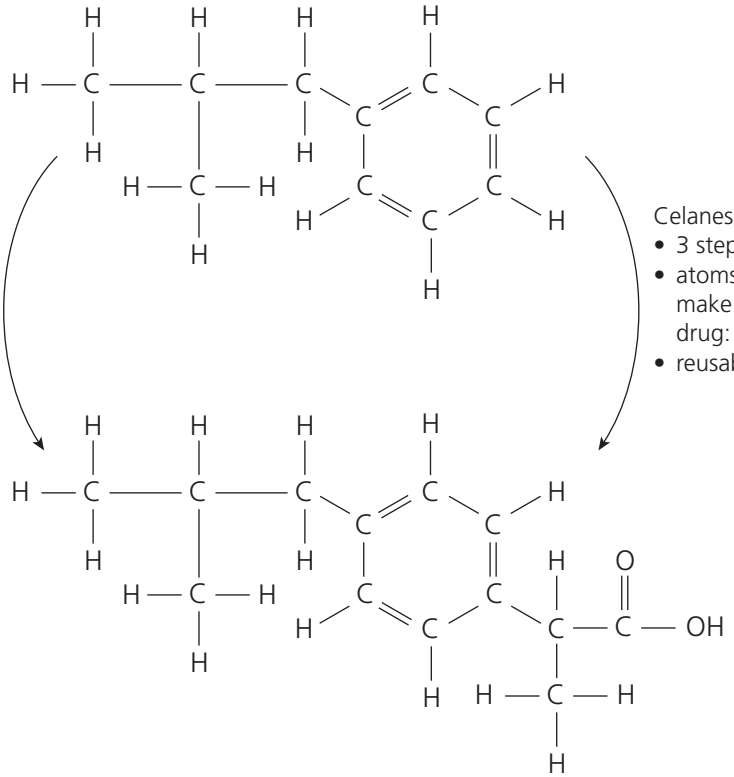
b) In *Method 2*, the reaction of copper metal with dilute nitric acid gives copper(II) nitrate, nitrogen monoxide and water. Write the chemical equation for this reaction.

c) Which of these methods would you recommend for the conversion of copper metal into copper(II) nitrate solution? Justify your answer with TWO reasons.

(HKCEE 2006)

7 The diagram below summarizes the two complex processes to make the pain reliever ibuprofen.
Refer to page T60.

Boots process
- 6 steps
- atoms in reactants to make one molecule of drug: 20C, 42H, 1N, 1O, 1Cl, 1Na
- non-reusable catalyst



Celanese process
- 3 steps
- atoms in reactants to make one molecule of drug: 15C, 22H, 4O
- reusable catalyst

a) What is the atom economy of the Boots process?
40.0%

b) What is the atom economy of the Celanese process?
77.4%

c) Suggest TWO reasons to explain why the Celanese process is 'greener' than the Boots process.

(Relative atomic masses: H = 1.0, C = 12.0, N = 14.0, O = 16.0, Na = 23.0, Cl = 35.5)

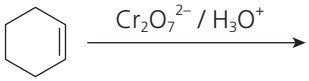
8 Any three of the following:
- Conditions — lower pressure / less separation required
- Feedstock — methanol can be obtained from a variety of sources
- Product — higher yield
- By-products — far fewer

8 The table below lists some information about two processes for the manufacture of acetic acid.

	Process	Conditions	Product
I	Oxidation of naphtha or butane	Temperature: 180 – 200 °C Pressure : 40 – 50 atm	The product contains about 35% acetic acid, together with a wide range of compounds, including propanone, methanoic acid, etc
II	Reaction of methanol with carbon monoxide	Temperature: 150 – 200 °C Pressure : 30 – 60 atm Catalyst: rhodium metal and iodide ion	About 98% of the methanol is converted to acetic acid.

From the green chemistry viewpoint, suggest THREE advantages of using the methanol / carbon monoxide process for the manufacture of acetic acid.

9 Hexanedioic acid, also known as adipic acid, is used in the manufacture of nylon-6,6. The acid is commonly synthesized from cyclohexene using method (I) or method (II) outlined below:

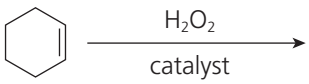
(I)  adipic acid

(II)  adipic acid

a) Draw the structure of adipic acid.

b) Both methods, (I) and (II), are considered as environmentally unfriendly. Explain.

c) Nowadays, some chemists recommend using method (III) below to synthesize adipic acid.

(III)  adipic acid

Suggest TWO advantages of using this method to synthesize adipic acid.

d) Nylon-6,6 is a polymer of adipic acid and hexane-1,6-diamine. Draw the repeating unit of nylon-6,6.

(HKASLE 2003)

10 Read the passage below and answer the questions that follow.

A greener bleach — hydrogen peroxide

A stain is a colour where you don't want it, but stain removal is not actually removal at all. The stain molecules are altered chemically instead.

The active ingredient in household chlorine bleach is sodium hypochlorite (NaOCl). The bleaching action is due to the action of the hypochlorite ion. However, the bleaching process often involves addition of chlorine atoms to the stain molecules. This can lead to the formation of hazardous by-products, such as dioxins.

Fortunately bleaches containing hydrogen peroxide are available. Whilst removing stains, the hydrogen peroxide decomposes to release highly reactive free radicals which oxidize other molecules.

Best of all, hydrogen peroxide does not contain any chlorine atoms and so produces no chlorine-containing pollutants.

Why not stop using chlorine bleach altogether and replace it with hydrogen peroxide alternatives? There are two reasons. First, molecules nearby may be affected by the reactive free radicals from hydrogen peroxide, leading to unwanted reactions. Second, the bleaching process requires higher temperatures and pressures, and longer reaction times than those required for chlorine bleach.

A research group in the USA has developed some molecules called tetraamido macrocyclic ligands or TAML for short. They are made from naturally occurring biochemicals and function as catalysts in the hydrogen peroxide bleaching process. Their presence allows hydrogen peroxide oxidation to proceed at much lower temperatures and pressures.

Bleaching with TAML-activated peroxide is an ideal example of green chemistry in action.

a) i) What reactants are used to prepare household chlorine bleach?
Chlorine and sodium hydroxide solution
ii) How are the reactants obtained?
Electrolysis of brine

b) The following label is displayed on the container of a chlorine bleach:

Keep out of the reach of children

Why should the bleach be kept out of the reach of children?

Suggest TWO potential hazards.
The bleach is toxic, irritating to eyes and skin.

c) Explain the meaning of the term 'free radical'.
A species containing an unpaired electron.

d) TAMLs can act as catalysts in the peroxide bleaching process.

i) Explain the meaning of the term 'catalyst'.
A catalyst is a substance which alters the rate of a reaction without itself undergoing any permanent chemical changes.

ii) Explain how catalysts increase the rate of a reaction.

10 d) ii) A catalyst provides an alternative reaction pathway with a lower activation energy.
As the activation energy is smaller, there is a larger proportion of molecules with energies equal to or greater than the new activation energy.
There is an increase in effective collision frequency. Hence the reaction goes faster.

e) From the perspective of green chemistry, describe THREE advantages of using TAML-activated peroxide for bleaching.
TAML are made from renewable resources.
TAML allow the hydrogen peroxide oxidation to proceed at much lower temperatures, and thus energy is saved.
No polluting end products are produced in the bleaching process.